A PROJECT REPORT ON

Digital Footprint Analysis for Suicide Risk Identification

SUBMITTED TO THE SAVITRIBAI PHULE UNIVERSITY, PUNE
IN THE PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE AWARD OF THE DEGREE

OF

BACHELOR OF ENGINEERING (Computer Engineering)

SUBMITTED BY

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2024-2025

CERTIFICATE

This is to certify that the project report entitled

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ACKNOWLEDGEMENT

It gives us great pleasure in presenting the project report on ***'Digital Footprint Analysis for Suicide Risk Identification'****.*

We would like to take this opportunity to thank ***Prof. Kimaya Urane*** *giving me all the help and guidance we needed. We are grateful to them for their kind support. Their valuable suggestions were very helpful.*

We are also grateful to ***Dr. G. V. Kale****, Head of Computer Engineering Department, PICT for her indispensable support, suggestions.*

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ABSTRACT

Suicide remains one of the most pressing and heartbreaking public health issues of our time. In 2021 alone, India witnessed a staggering 1,64,033 reported suicide cases — a number that underscores the urgent need for timely detection and effective prevention strategies. Behind every statistic is a life lost, a family devastated, and a conversation that perhaps never happened.

With the widespread use of the internet and the ever-increasing number of digital footprints we leave behind, there is an opportunity to harness technology for good. This project explores a proactive approach to suicide prevention by using digital behaviour as an early warning system. By tracking and analysing a person's online browsing patterns through a custom-built Chrome extension, this system attempts to detect signs of depression and suicidal ideation. The backbone of this detection process is the Suicide and Depression Detection Dataset, which trains the system to recognize potentially harmful or concerning behaviour in a user's online activity.

The ultimate goal is not surveillance, but support — to serve as a quiet observer that can raise a flag before it is too late. This tool can prove especially useful for parents, educators, and mental health professionals who may otherwise miss subtle signs of distress. By enabling early intervention, it offers a chance to start meaningful conversations and provide help when it is needed most.

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Chapter 1

Introduction

1.1 Overview

Natural Language Processing (NLP) is a subfield of artificial intelligence and computational linguistics that focuses on enabling computers to understand and process human language. NLP seeks to bridge the gap between human communication and computer understanding, by enabling computers to analyze and understand the complexities of human language in a way that is meaningful and useful. NLP is used to detect the texts present in the websites visited and identify websites which could induce thoughts of depression and suicide.

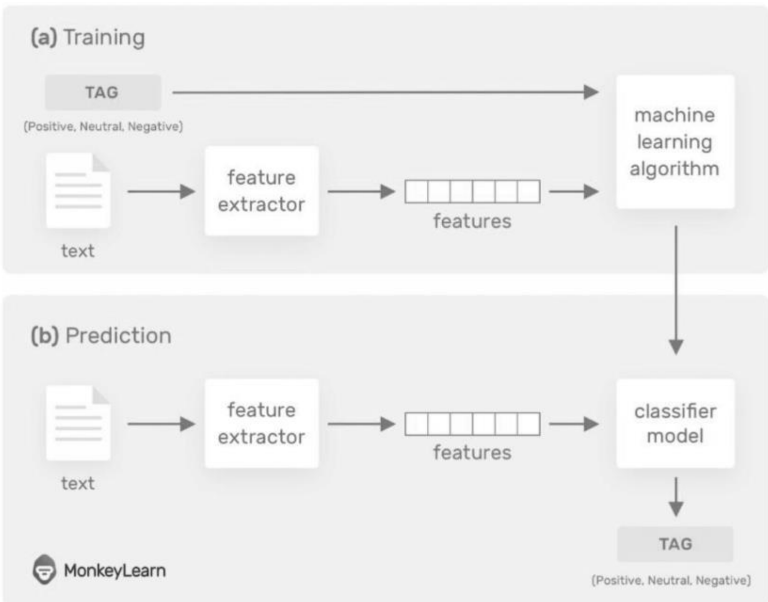


Fig 1.1: NLP Working

1.2 Objectives

Tracking browsing provides a faster and more efficient way of identifying potential suicidal behaviour as traditional methods of suicide detection and prevention rely on direct and periodic observation and assessment by mental health professionals which has been proven to be inadequate. Our main objectives are:

i. Early Detection: One of the main objectives of a suicide detection system is to identify individuals at risk of suicide as early as possible by tracking their browsing.

ii. Real-time Monitoring: Suicide detection systems aim to provide real-time browsing updates of individuals at risk of suicide.

iii. Integration with Support Systems: Suicide detection systems can be integrated with existing support systems, such as mental health services, crisis hotlines, or emergency response protocols, to facilitate timely intervention and support.

iv. Accurate Identification: Another objective of a suicide detection system is to accurately identify individuals who may be at risk of suicide, while minimizing false positives and negatives.

1.3 Problem Definition

In today's fast-paced world, emotional well-being often goes unnoticed, leading to increased stress and mental health issues. Detecting human emotions using facial expressions can significantly enhance communication, therapy, and human-computer interaction. However, achieving real-time and accurate emotion recognition from facial cues remains a technical challenge. This project aims to develop a system that accurately identifies and classifies human emotions using image-based facial analysis with machine learning techniques.

1.4 Project Scope & Limitations

i. Emotion Detection Through Facial Analysis: Develop a system capable of recognizing human emotions such as happiness, sadness, anger, fear, and neutrality through facial expressions using computer vision and machine learning techniques.

ii. Integration with Chrome Extension: Implement a Chrome browser extension that passively monitors user behavior and browsing patterns, helping to identify digital signs of depression or suicidal tendencies.

iii. Use of Suicide and Depression Detection Dataset: Train and evaluate the model using a specialized dataset that includes facial emotion data and behavioral indicators related to mental health issues.

iv. Real-Time Monitoring and Alerts: Enable real-time emotion detection and risk assessment, providing timely alerts or recommendations for intervention to concerned authorities such as parents, teachers, or counselors.

v. Application in Educational and Domestic Environments: Deploy the system in schools, colleges, and homes to monitor students or young individuals for early signs of emotional distress, offering an additional layer of support.

vi. Enhancing Mental Health Awareness: Promote mental health awareness by encouraging open conversations and offering technological assistance to identify and support at-risk individuals.

vii. Scalability and Future Expansion: Design the system to be scalable, with potential future upgrades to include emotion recognition from voice inputs, text-based sentiment analysis, and integration with wearable devices for more accurate assessments.

1.5 Methodologies of Problem solving

1. Problem Definition and Understanding
   - Goal: Clearly define the problem, understand the requirements, constraints, and desired outcomes.
   - Example: "Detect early signs of depression or suicidal tendencies by analyzing facial expressions and browsing behavior in real time."
2. Data Collection and Preprocessing
   - Goal: Gather relevant, high-quality data and clean it for analysis.
   - Steps: Handle missing values, remove outliers, normalize or scale features, extract useful indicators or features.
3. Model Selection and Design
   - Goal: Choose or design the right model architecture suited for the problem.
   - Steps: Try various models (e.g., LSTM, DistilBERT) and select based on task complexity and data characteristics.
   - Example: Use LSTM for sequence modeling, possibly extend to Seq2Seq for multi-day forecasts.
4. Training, Validation, and Evaluation
   - Goal: Train the model with appropriate hyperparameters and evaluate it using performance metrics.
   - Steps: Split data into training/validation, monitor loss and metrics like MAE, RMSE, R2. Avoid overfitting.
5. Iteration and Optimization
   - Goal: Continuously refine the solution based on feedback, results, and edge cases.
   - Steps: Analyze where the model fails, tune hyperparameters, enrich features, or change modeling strategies.
   - Example: If predictions deviate at the end, try longer input sequences, re-train on recent data, or adjust forecast strategy.

Chapter 2

Literature Survey

2.1 Introduction

In recent years, there has been a notable increase in the body of research on the automatic classification and detection of suicidal tendencies using machine learning (ML) and natural language processing (NLP) techniques. The potential of natural language processing (NLP) algorithms to analyse text data—especially from social media platforms—to identify people who may be at risk of suicidal thoughts and behaviours has been the subject of numerous studies. Mulholland and Quinn (2012) demonstrated promising classification rates in their groundbreaking application of natural language processing (NLP) to predict musicians' suicide risk from song lyrics. Building on this basis, later studies have demonstrated the adaptability of NLP techniques in recognizing psychological conditions, including suicide thoughts, from verbal expressions, as demonstrated by Young et al.'s (2023) study. By using Transformer models to identify suicidal ideation in social media posts, Haque et al. (2020) advanced this field and demonstrated the potency of deep learning architectures. The focus was widened to include machine learning analysis of Twitter data by Roy et al. (2020) and Tadesse et al. (2019), who emphasized the potential of algorithms in predicting future risks of suicidal thoughts. Arowosegbe et al. (2023), focusing on social media forums, emphasized the value of natural language processing (NLP) in identifying and averting suicidal ideation and provided information on low-cost substitutes for suicide prevention frameworks. Furthermore, studies like those by Mo et al. (2022) and Haque et al. (2022) have investigated the psychological mechanisms underlying suicidal behaviour and performed comparative analyses, respectively, which have added to our understanding of the intricate factors at play. We hope to synthesize current knowledge, highlight research gaps, and suggest future directions for this important field of mental health research through this review of the literature.

2.2 Literature Review

2.2.1. Suicidal Tendencies: The Automatic Classification of Suicidal and Non-Suicidal Lyricists Using NLP.

Authors: Matthew Mulholland, Joanne Quinn
Year: 2012
Observation:
In this paper, natural language processing was used to predict the likelihood that a musician will commit or has committed suicide? To explore this idea, a corpus of songs was built that includes a development set, a training set, and a test set, all consisting of different lyricists. Various vocabulary and syntactic features were then calculated to create a suicide/non-suicide song classifier. The features were input into the Weka machine learning suite and tested with an array of algorithms. The experiment resulted in up to a 70.6% classification rate with the Simple Cart algorithm, a 12.8% increase over the majority-class baseline. The findings from the research suggested that syntactic and vocabulary features are useful indicators of the likelihood that a lyricist will commit suicide.

2.2.2 A Review of natural language processing in the Identification of suicidal behaviour.

Authors: John Young, Steven Bishop a, Carolyn Humphrey a, Jeffrey M. Pavlacic b.
Year: 2023
Observation:
Natural language processing (NLP) approaches offer powerful, flexible methods of identifying various psychological conditions from free-form verbal expression in both written and spoken form. These tools may be particularly useful for suicide risk detection and potentially easing human suffering on a global scale. The degree of their accuracy in discerning suicidal thoughts or behaviours, however, is currently unknown. Results: Studies were diverse in terms of performance statistics reported, with inconsistencies across fields and publications. Regardless, all metrics suggested strong performance of NLP methods tested, which were heavily slanted toward accurate identification of positive cases (i.e., those at elevated risk for suicidal thoughts or behaviours).

Limitations: Reports were non-standardized across studies, necessitating a non-optimal combination of performance statistics through non-weight averages. Small sample sizes and biased test construction samples were also notable in some studies, and the generalizability of study results was variable.
Conclusions: Use of these strategies in applied environments may enhance early identification and prevention, and future work to understand their dissemination and implementation is recommended.

2.2.3 A Transformer-Based Approach to Detect Suicidal Ideation Using Pre-Trained Language Models.

Authors: Farsheed Haque, Ragib Un Nur, Shaeekh Al Jahan, Zarar Mahmud, and Faisal Muhammad Shah
Year: 2020
Observation:
Detection of Suicidal Ideation in social media has gained special attention in recent years. Different mental health issues like depression, frustration, hopelessness, etc directly or indirectly influence suicidal thoughts. Machine Learning and Deep Learning are playing a vital role in this area to automatize Suicidal Ideation detection. In our study, we wanted to take this trend to the next level and proposed a new detection approach with the help of the Transformer models in the language domain. Basically, with Transformers, we wanted to analyse raw social media posts and classify the indication of suicidal ideation in them. First, we applied BiLSTM(Bidirectional Long Short-Term Memory) and from there took a jump to Transformer models like BERT (Bidirectional Encoder Representations from Transformers), ALBERT, ROBERTa, and XLNET - a complete robust Transformer model-based study. The main advantage of Transformer models is having pretrained language models. Because of this, these models usually perform better. We found they work far better than conventional Deep Learning architecture like Bi-LSTM in our work.

2.2.4 A machine learning approach predicts future risk of suicidal ideation from social media data.

Authors: Arunima Roy, Katerina Nikolitch, Rachel McGinn, Safiya Jinnah, William Klement and Zachary A. Kaminsky
Year: 2020
Observation:
Machine learning analysis of social media data represents a promising way to capture longitudinal environmental influences contributing to individual risk for suicidal thoughts and behaviours. The objective was to

generate an algorithm termed "Suicide Artificial Intelligence Prediction Heuristic (SAIPH)" capable of predicting future risks to suicidal thoughts by analysing publicly available Twitter data. The author trained a series of neural networks on Twitter data queried against suicide associated psychological constructs including burden, stress, loneliness, hopelessness, insomnia, depression, and anxiety. The research validated the model using regionally obtained Twitter data and observed significant associations of algorithm SI scores with county-wide suicide death rates across 16 days in August and in October 2019, most significantly in younger individuals. Algorithmic approaches like SAIPH have the potential to identify individual future SI risk and could be easily adapted as clinical decision tools aiding suicide screening and risk monitoring using available technologies.

2.2.5 Machine Learning for suicidal ideation identification: A systematic literature review.

Authors: Wesllei Felipe Heckler, Juliano Varella de Carvalho, Jorge Luis Victória Barbosa
Year: 2022
<u>Observation</u>:
Suicidal ideation is the first stage in the risk scale, being a potential gate for suicide prevention. Machine learning emerged as a promising tool for helping in preventing suicide through the identification of individuals at risk. Therefore, this paper presents how machine learning can help in suicidal ideation identification. The author proposed a taxonomy of machine learning techniques explored in this area and a taxonomy for highlighting the current research challenges. In a general way, studies explored data from social media and performed a text analysis to investigate suicidal tendencies in the individuals' language. Moreover, deep learning models seem to be a tendency in this area nowadays. Future studies in suicidal ideation should investigate generic and proactive models that do not depend on users' self-report.

2.2.6 Detection of Suicide Ideation in Social Media Forums Using Deep Learning

Authors: Michael Mesfin Tadesse, Hongfei Lin, Bo Xu and Liang Yang
Year: 2019
<u>Observation</u>:
Suicide ideation expressed in social media has an impact on language usage. Many at-risk individuals use social forum platforms to discuss their

problems or get access to information on similar tasks. The key objective of our study is to present ongoing work on automatic recognition of suicidal posts. The author addressed the early detection of suicide ideation through deep learning and machine learning-based classification approaches applied to Reddit social media. For such purpose, an LSTM-CNN combined model was employed to evaluate and compare to other classification models. The experiment shows the combined neural network architecture with word embedding techniques can achieve the best relevance classification results. Additionally, the results support the strength and ability of deep learning architectures to build an effective model for a suicide risk assessment in various text classification tasks.

2.2.7 Application of Natural Language Processing (NLP) in Detecting and Preventing Suicide Ideation

Authors: Abayomi Arowosegbe, Tope Oyelade, and Paul B. Tchounwou
Year: 2023
Observation:
Around a million people are reported to die by suicide every year, and due to the stigma associated with the nature of the death, this figure is usually assumed to be an underestimate. Machine learning and artificial intelligence such as natural language processing have the potential to become a major technique for the detection, diagnosis, and treatment of people.
Methods: PubMed, EMBASE, MEDLINE, PsycINFO, and Global Health databases were searched for studies that reported the use of NLP for suicide ideation or self-harm.
Result: The preliminary search of 5 databases generated 387 results. Removal of duplicates resulted in 158 potentially suitable studies.
Conclusions: The use of AI&ML opens new avenues for considerably guiding risk prediction and advancing suicide prevention frameworks. The review's analysis of the included research revealed that the use of NLP may result in low-cost and effective alternatives to existing resource-intensive methods of suicide prevention.

2.2.8 A Comparative Analysis of Suicidal Ideation Detection Using NLP, Machine and Deep Learning

Authors: Rezaul Haque, Naimul Islam, Maidul Islam, and Md Manjurul Ahsan
Year: 2022

Observation:
Social networks are essential resources to obtain information about people's opinions and feelings towards various issues as they share their views with their friends and family. Suicidal ideation detection via online social network analysis has emerged as an essential research topic with significant difficulties in the fields of NLP and psychology in recent years. With the proper exploitation of the information in social media, the complicated early symptoms of suicidal ideations can be discovered and hence, it can save many lives. This study offers a comparative analysis of multiple machine learning and deep learning models to identify suicidal thoughts from the social media platform Twitter. The author applied text pre-processing and feature extraction approaches such as Count Vectorizer and word embedding, and trained several machine learning and deep learning models for such a goal. Experiments were conducted on a dataset of 49,178 instances retrieved from live tweets by 18 suicidal and non-suicidal keywords using Python Tweepy API. The experimental findings reveal that the RF model can achieve the highest classification score among machine learning algorithms, with an accuracy of 93% and an F1 score of 0.92.

2.2.9 Exploring the Suicide Mechanism Path of High-Suicide-Risk Adolescents—Based on Weibo Text Analysis

Authors: Liuling Mo,He Li, and Tingshao Zhu
Year: 2022
Observation:
This study explored the relationship between psychological pain, hopelessness, and suicide stages of high-suicide-risk adolescents. Qualitative analysis showed that 36.2% of high suicide-risk adolescents suffered from mental illness, and depression accounted for 76.3% of all mental illnesses. The mediating effect analysis showed that hopelessness played a complete mediating role between psychological pain and suicide stages, and was significantly negatively correlated with suicide stages. The suicide mechanism path in high-suicide-risk adolescents is psychological pain and hopelessness suicide stages, indicating that psychological pain mainly affects suicide risk through hopelessness. Adolescents who are later in the suicide stages have fewer expressions of hopelessness.

Chapter 3

Software Requirements Specification

3.1 Assumptions and Dependencies

1. Assumptions
   - Clean Historical Data Available
     It is assumed that the Suicide and Depression Detection dataset is pre-cleaned and contains only text.
   - Text-Based Features Are Sufficient
     It is assumed that linguistic features such as sentiment, emotion cues, and suicidal ideation patterns in text are sufficient for accurate detection.
   - User consent and awareness
     It is assumed that users will access the detection system through a web interface, such as a Chrome extension.
2. Dependencies
   - Libraries and Frameworks
     The system depends on external Python libraries such as:
     - Natural Language Processing: NLTK, spaCy, TextBlob
     - Deep Learning: TensorFlow/Keras or PyTorch
     - Preprocessing: Pandas, NumPy
     - Visualization: Matplotlib, Seaborn
     - Transformers: HuggingFace Transformers (e.g., BERT)

- Frontend-Backend Integration
  The system relies on smooth interaction between a browser extension frontend and Python-based backend via Flask/Django.
- Dataset Format
  Text data is stored in CSV/JSON format with columns like text, label, and possibly timestamp.
- Hardware/Runtime Environment
  The system depends on sufficient computational resources (CPU/GPU and RAM) to train or run prediction models smoothly without lag.

3.2 Functional Requirements

The following are the functional requirements for the proposed system:

3.2.1 System Feature 1: Suicide Risk Detection (Functional Requirement)

- **Description:** The system detects emotional states and potential suicide risk based on user-generated text input.
- **Functionalities:**
  - Monitor and analyze user text from Chrome browser or input field.
  - Classify text as **Neutral**, **Depressive**, or **Suicidal**.
  - Alert user or responsible authority if suicidal text is detected.
  - Optionally store flagged text for later analysis.
- **Input:** Raw text content from user input or web interactions.
- **Output:** Risk classification label and alert.

3.2.2 System Feature 2: User Interface and Visualization (Functional Requirement)

- **Description:** A Chrome extension or web interface provides a minimal and user-friendly dashboard for viewing results and history.
- **Functionalities:**
  - Text box or passive monitoring through browser.
  - Display of classification result with color-coded risk levels
  - Graph of emotional trend over time (optional).

  - Option to export detection history as CSV.
- **Input:** User interaction through Chrome extension or web UI
- **Output:** Risk label display, trend visualization, downloadable report.

3.3 External Interface Requirements

The following are the external interface requirements for the proposed system:

3.3.1 User Interfaces

- Simple, accessible Chrome extension or browser-based UI.
- Users will be able to:
  - Input field or passive scanning toggle.
  - Risk level indicator.
  - Historical trend graph.
  - CSV export option.

3.3.2 Hardware Interfaces

- No specialized hardware is required.
- The system can run on:
  - Standard PCs or laptops with internet access.
  - Server environments for hosted versions (GPU for faster training).
- Recommended minimum:
  - 4 GB RAM, Dual-core CPU, 5 GB storage.

3.3.3 Software Interfaces

- **Operating System:** Compatible with Windows, Linux, or macOS.
- **Backend Environment:** Python 3.8+, Flask/Django (for server-side logic).
- **Libraries:** spaCy, NLTK, TensorFlow, HuggingFace Transformers
- **Frontend Stack:** HTML/CSS/JavaScript
- **Browser Support:** Chrome, Firefox

3.3.4 Communication Interfaces

- **Local Deployment**
  - HTTP-based interaction between frontend and backend.
  - Localhost or server IP
- **Data Input/Output**
  - Text input via Chrome extension
  - No external APIs are currently used

3.4 Nonfunctional Requirements

3.4.1 Performance Requirements

- The system shall provide predictions within 1-2 seconds of user input under normal load conditions.
- The UI shall respond to user interactions within 1 second.
- Should handle concurrent analysis for multiple users in web deployment.

3.4.2 Safety Requirements

- Text is handled securely with local storage unless user opts in.
- System should prevent data loss and gracefully handle errors.

3.4.3 Security Requirements

- The system must validate all inputs to prevent injection attacks or malformed data uploads.
- File uploads must be restricted to CSV/JSON format and scanned to prevent malicious execution.
- For hosted deployments:
  - HTTPS shall be enforced to secure data transmission.
  - Basic authentication and role-based access control can be implemented if user accounts are supported.
  - No sensitive user data is collected or stored by default.

3.4.4 Software Quality Attributes

Quality Attributes

- **Usability**: Intuitive Chrome extension and visual feedback.
- **Maintainability**: Modular code (model, UI, server separated).
- **Portability**: Cross-platform compatibility.
- **Reliability**: Robust performance with accurate classification.
- **Scalability**: Robust performance with accurate classification.

3.5 System Requirements

3.5.1 Database Requirements

The system does not rely on a traditional database like MySQL or MongoDB. Instead, it uses structured CSV/JSON files to manage historical data. Optionally, a lightweight database like SQLite or Firebase can be introduced for logging prediction history, managing user preferences, or future scalability.

3.5.2 Software Requirements (Platform Choice)

The backend of the system is developed in Python 3.10 or higher. Libraries and frameworks used include Pandas, NumPy, Matplotlib, TensorFlow for model implementation, and Scikit-learn for data preprocessing. Flask or

Django is used for the web application backend, while the frontend may optionally use simple HTML/CSS or ReactJS. Development is typically done in environments like Jupyter Notebook, VS Code, or Py- Charm. For deployment, platforms such as Localhost, Heroku or PythonAnywhere can be used.

3.5.3 Hardware Requirements

The minimum hardware required for running the application includes a dual-core CPU, 4 GB of RAM, and at least 5 GB of free disk space. However, for smoother execution and training of models, it is recommended to use a quad-core processor, 8 GB or more of RAM, and an SSD with at least 10 GB of free space. While a GPU is not mandatory, having an NVIDIA GPU significantly speeds up model training, especially for deep learning tasks.

3.6 Analysis Models: SDLC Model to be applied

The project follows the **Iterative Incremental Model** of the Software Development Life Cycle (SDLC). **Justification:**

1. **Modular Approach:** Risk detection and UI components are developed separately.
2. **Frequent Feedback:** Allows tuning based on model accuracy and UI experience.
3. **Flexibility:** Easy to expand with new emotional labels or multilingual support.
4. **Parallel Development:** Backend (ML model) and frontend (Chrome UI) evolve independently

Chapter 4

System Design

4.1 System Architecture

Figure 4.1 illustrates the system architecture used for prediction in this project.

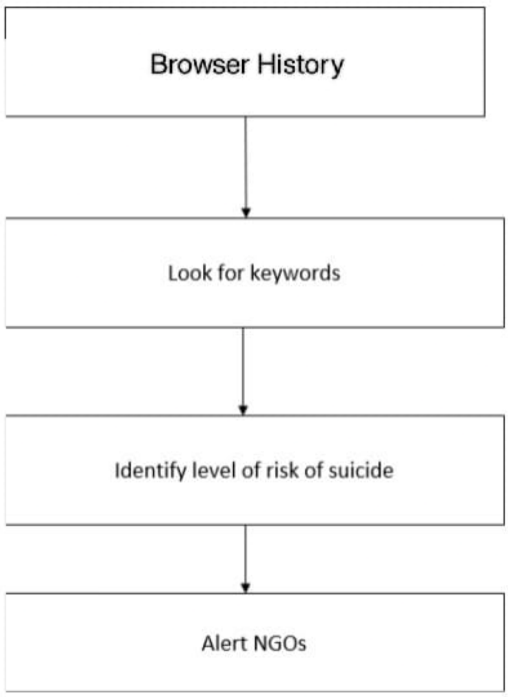


Figure 4.1: System Architecture

Data collection: The system would have to gather textual data from several places, including internet forums, chat logs, and social networking sites. The system might work with crisis hotlines or mental health groups to get data from their chat logs or counselling sessions, or it could use the APIs that social media sites provide to gather public posts.

Data preprocessing: To clean and normalize the text data, stop words, punctuation, and other unnecessary information must be removed. It is also possible to use NLP techniques like Bag of Words, TF-IDF, or Word Embeddings to tokenize, stem, and vectorize the data.

Feature Extraction: From the text data, the system could then extract pertinent features like sentiment analysis, topic modelling, or linguistic features like word frequency or syntax. These characteristics could be utilized to train models that anticipate the risk factors for suicide or the probability of suicidal thoughts.

Model development: To classify text data into suicidal and non-suicidal categories, the system could make use of a variety of machine learning models, including logistic regression, support vector machines, and deep learning model like Transformer. Cross-validation techniques could be employed to validate the models after they have been trained on a labelled dataset.

Depression classification: Based on the users' text data, the system could then use the trained models to determine each user's risk level. If someone is judged to be at high risk of suicide, the system may send an automated alert with suggested resources or interventions to crisis hotlines or mental health specialists.

Evaluation and enhancement: To guarantee precision and efficiency, the system must be routinely assessed and modified. Metrics including precision, recall, F1 score, and ROC curve analysis could be used to assess the system. Over time, the system could also be improved with the help of user feedback data.

4.2 Mathematical Model

Figure 4.2 represents the architecture of the DistilBERT model used for time-series forecasting.

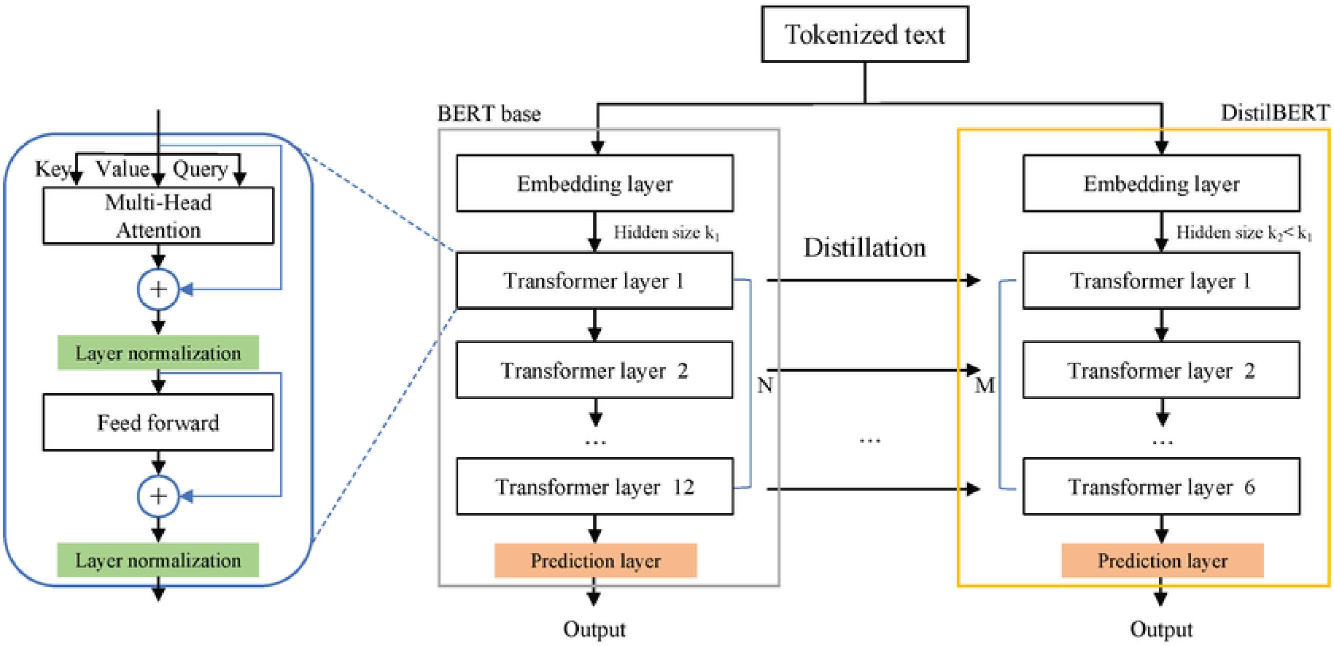


Figure 4.2: DistilBERT Architecture

4.3 Data Flow Diagrams

Figure 4.3 description of each major software function is presented, along with data flow (structured analysis) or class hierarchy (Analysis Class diagram with class description for object-oriented system).

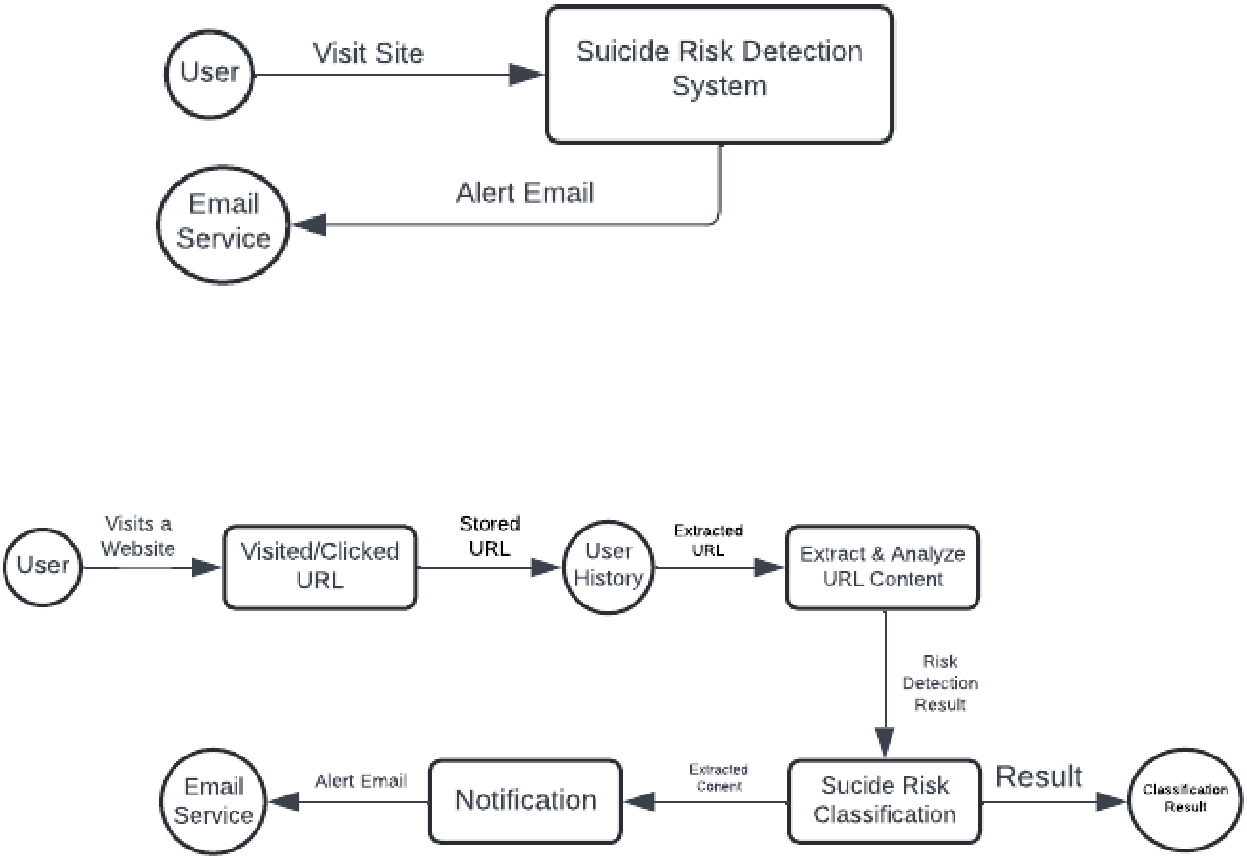


Figure 4.3: Data Flow Diagram

4.4 Entity Relationship Diagram

Figure 4.4 illustrates the Entity Relationship Diagram (ERD) for the proposed system. It outlines the main entities involved and their relationships.

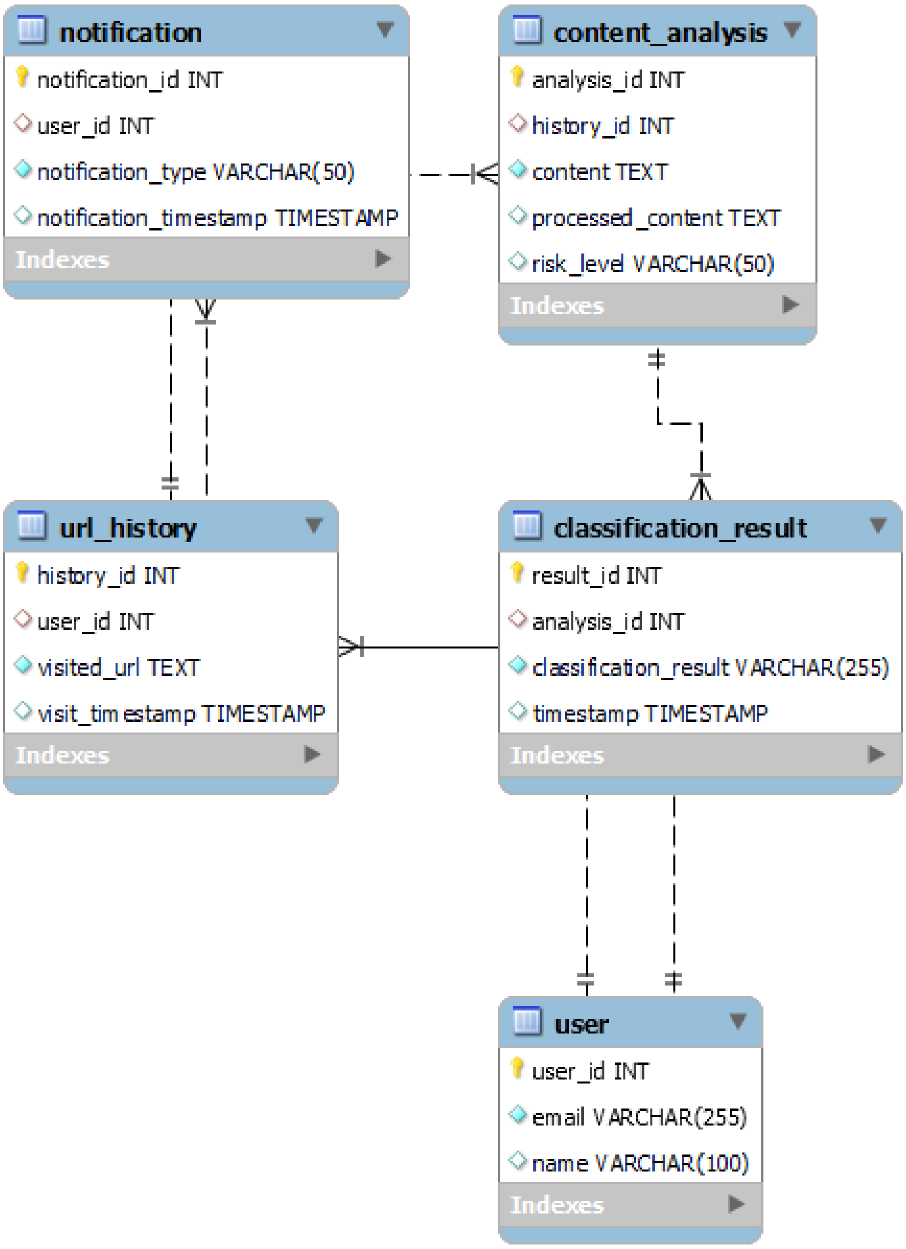


Figure 4.4: Entity Relationship Diagram

4.5 UML Diagram

4.5.1 Class Diagram

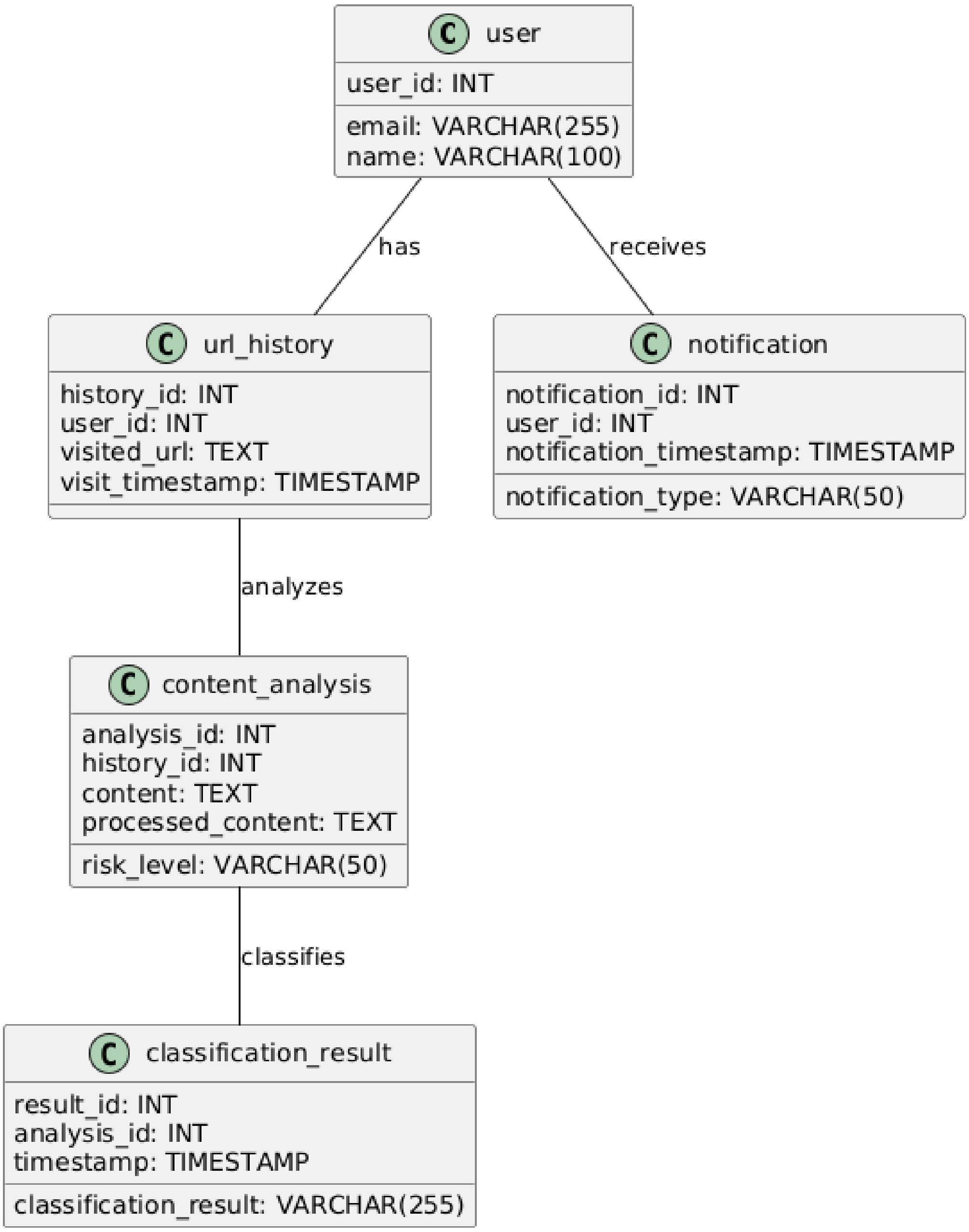


Figure 4.5: Class Diagram

4.5.2 Use Case Diagram

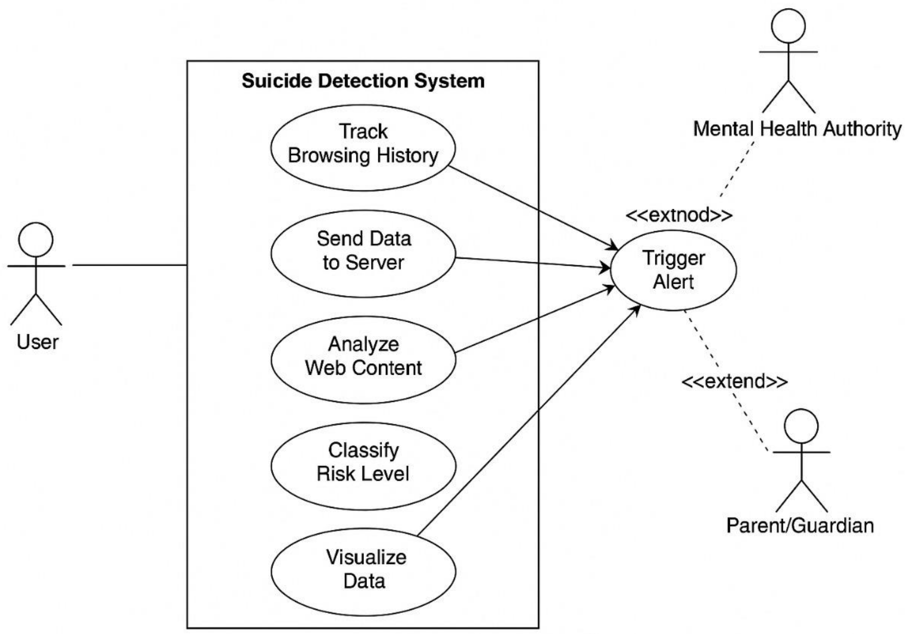


Figure 4.6: Use Case Diagram

4.6 Activity Diagram

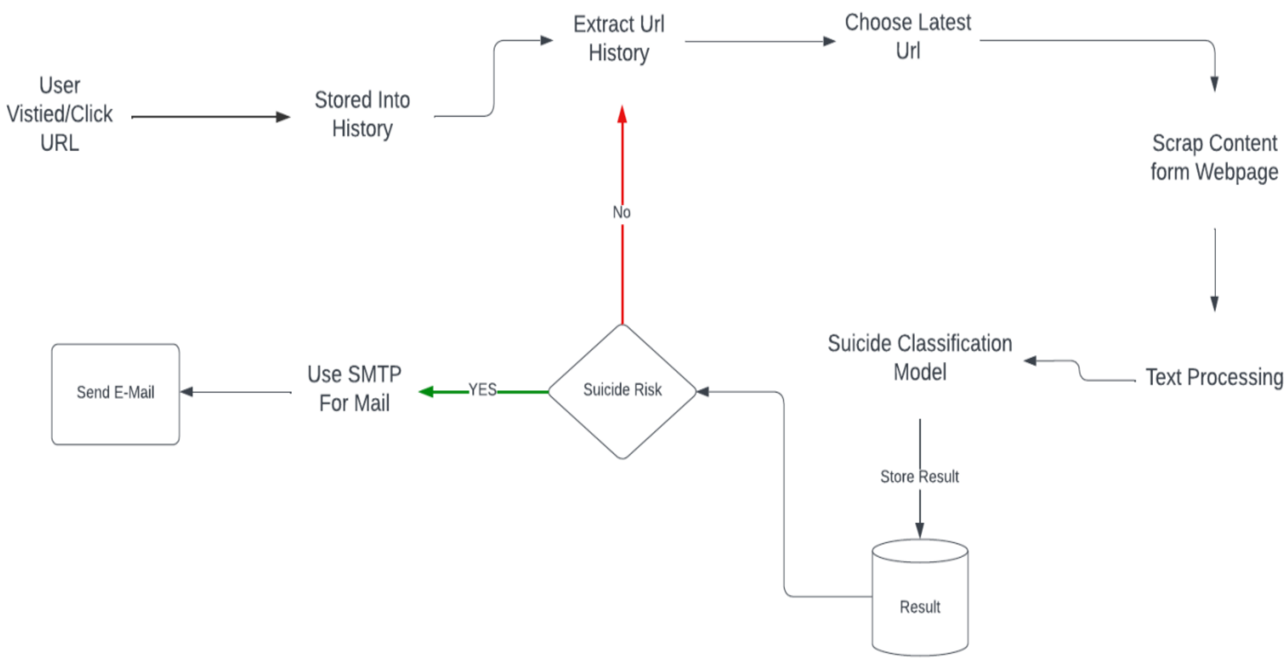


Figure 4.7: Activity Diagram

4.7 Sequence Diagram

Figure 4.8 represents the Sequence Diagram of the proposed system. This diagram models the sequential interaction between different system components over time.

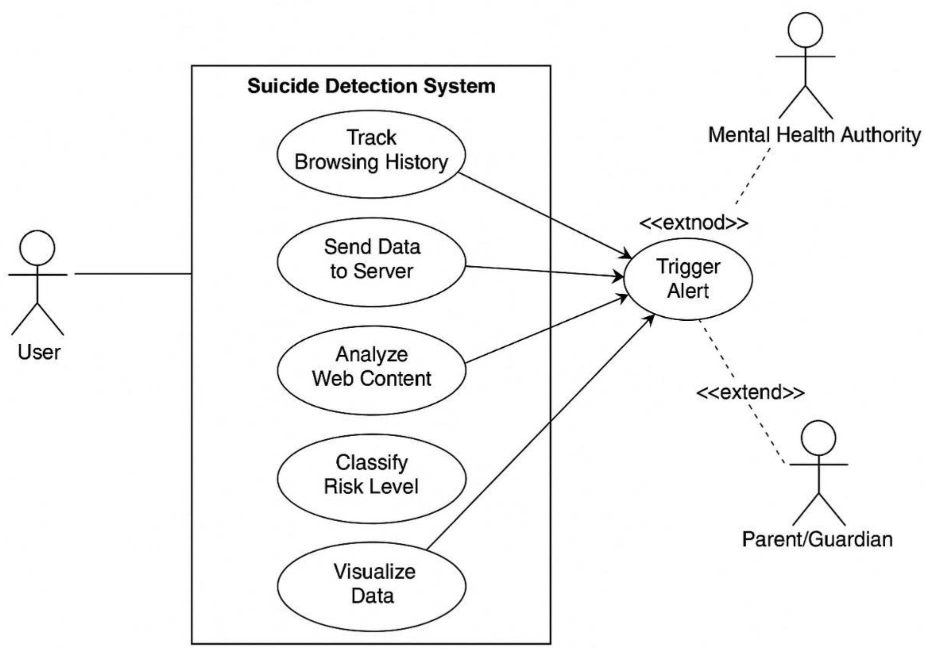


Figure 4.8: Sequence Diagram

Chapter 5

Project Plan

5.1 Project Estimate

5.1.1 Reconciled Estimates

- **Project Planning and Preparation (2 weeks):**
  Initial discussions, requirement gathering, objective definition, and tool selection for a text-based suicide and emotion detection system.
- **Studying Scope and Feasibility (1 week):**
  Evaluating the feasibility of using only text data (e.g., social media comments, chat logs) for emotion classification and suicide risk detection. Exploring datasets like Reddit mental health datasets.
- **Data Acquisition and Preprocessing (1 month):**
  Acquiring labeled datasets related to mental health, depression, and suicide. Preprocessing includes text cleaning, tokenization, stop-word removal, lemmatization, and vectorization using methods like TF-IDF or BERT embeddings.
- **Model Development and Training (2.5 months):**
  - **Text Feature Engineering (2 weeks):** Extracting sentiment scores, emotional tone (using tools like VADER/TextBlob), and linguistic features from input text.
  - **Model Development (6 weeks):** Designing and fine-tuning classification models such as Logistic Regression and transformer-based models (e.g., BERT) to detect suicidal intent and depression from text.

- **Model Evaluation and Optimization (1.5 months):**
  Evaluating models using metrics like Precision, Recall, F1-score, and confusion matrix. Handling class imbalance using SMOTE or weighted loss. Hyperparameter tuning via grid search or Bayesian optimization.
- **Software Integration and Chrome Extension Development (1 month):**
  - **Frontend Development (2 weeks):**
    Building a Chrome extension or simple web interface for real-time text input and result visualization.
  - **Backend Integration (2 weeks):**
    Developing REST APIs to connect frontend input with the trained model for prediction.
- **Testing and Quality Assurance (2 weeks):**
  Performing unit testing, model validation, frontend-backend integration testing, and verifying alert mechanisms in high-risk cases.
- **Documentation and Presentation (2 weeks):**
  Preparing detailed technical documentation, user guide, flow diagrams, and final presentation deck for demonstration.
- **Final Review (1 week):**
  Collecting feedback from users to apply final improvements and preparing for deployment or submission.

5.1.2 Project Resources

- **Software Development:** Designed, developed, and implemented the core backend modules required for the project, including data acquisition, preprocessing, feature engineering, model training, evaluation, and prediction pipelines. Used Python and essential libraries such as Pandas for data manipulation, NumPy for numerical computation, and scikit-learn for preprocessing and baseline model development.
- **Deep Learning Model Development:** Built and trained various NLP models including Logistic Regression (as baseline), DistilBERT for sequence learning and for emotion and suicide risk classification. Employed dropout, early stopping, and learning rate schedulers to optimize performance and avoid overfitting.

- **Data Sources and Tools:** Used publicly available mental health datasets from Reddit, Kaggle, and the Suicide and Depression Detection dataset. Employed VADER and TextBlob for sentiment scoring and feature augmentation. Data augmentation techniques were used to balance minority classes.
- **User Interface and Backend Development:** Developed a simple Chrome Extension or web UI using HTML, CSS, and JavaScript (optionally React) for user interaction. Flask was used to handle API routing and connect frontend input to the trained backend model for real-time inference.
- **Collaboration and Development Tools:** Used GitHub for version control, collaboration, and issue tracking. Used Google Colab and Jupyter Notebooks for model experimentation and team prototyping. Conducted regular team meetings via Google Meet to align milestones and resolve blockers.

5.2 Risk Management

5.2.1 Risk Identification

During the risk identification process for our project, we identified various risks:

- **Technical Risks:**
  - **Model Generalization**: Risk of inaccurate predictions due to biased or limited data was mitigated by using diverse datasets and applying stratified sampling.
  - **Model Complexity**: Fine-tuning transformer models like BERT posed challenges, addressed through staged experimentation and monitoring.
  - **Data Quality**: Noise, mislabeled samples, or imbalance in mental health text datasets were handled via rigorous text cleaning and class balancing techniques.
  - **Data Quality**: Noise, mislabeled samples, or imbalance in mental health text datasets were handled via rigorous text cleaning and class balancing techniques.

- **Ethical and Privacy Risks:**
  - **Sensitive Data Handling**: Since the system deals with potentially personal or mental health–related text, ethical risks were addressed by anonymizing data and using only public or consented datasets.
  - **Misclassification Consequences**: False positives or negatives could lead to incorrect alerts. Risk was addressed with high precision thresholds and confidence-based alert triggering.
- **Operational Risks:**
  - **Scope Creep**: Controlled by well-defined deliverables and adherence to a project plan.
  - **Team Coordination**: Risk from inconsistent development efforts was mitigated through weekly meetings, documentation, and role assignments.
  - **Dependency on External Libraries/APIs**: Managed by using stable versions and providing fallback models for offline operation.

5.2.2 Risk Analysis

After identifying risks, a detailed analysis was performed to assess impact and define mitigation steps. This analysis supported a proactive strategy, ensuring successful project completion.

- **Technical Risks**
  - **Overfitting & Underfitting**: Controlled using dropout, early stopping, cross-validation, and continuous evaluation with new data samples.
  - **Imbalanced Classes**: Managed by applying oversampling techniques (e.g., SMOTE) and using weighted loss functions.
- **Ethical and Privacy Risks**
  - **Imbalanced Classes**: Managed by applying oversampling techniques (e.g., SMOTE) and using weighted loss functions.
  - **Imbalanced Classes**: Managed by applying oversampling techniques (e.g., SMOTE) and using weighted loss functions.
- **Operational Risks**
  - **Imbalanced Classes**: Managed by applying oversampling techniques (e.g., SMOTE) and using weighted loss functions.
  - **Tool Failures**: Backup models, version control, and cloud backups were prepared to avoid workflow disruption.

- **Security Risks**
  - **System Vulnerabilities**: Addressed through basic input sanitization, secure API endpoints (e.g., via HTTPS), and updated dependencies.
  - **User Data Protection:** Data was processed locally or in a secure environment with no external transmission unless necessary.

5.2.3 Overview of Risk Mitigation, Monitoring and Management

- **Technical Risk Mitigation**
  - Conducted regular testing, model evaluations, and refactoring to ensure consistent accuracy and stability.
- **Ethical Risk Mitigation**
  - Implemented anonymization protocols and used only approved datasets. Deployed a feedback mechanism to flag false results.
- **Operational Risk Mitigation**
  - Applied agile methods with sprints, clear scope, and versioned task lists. Used GitHub for transparency and issue tracking.
- **Risk Monitoring**
  - Weekly risk assessments were conducted with logs maintained for any newly surfaced issues, addressed in real-time.
- **Risk Management**
  - A buffer time of 10% was built into each phase for risk absorption. Resource reallocation was pre-planned for critical delays or gaps.

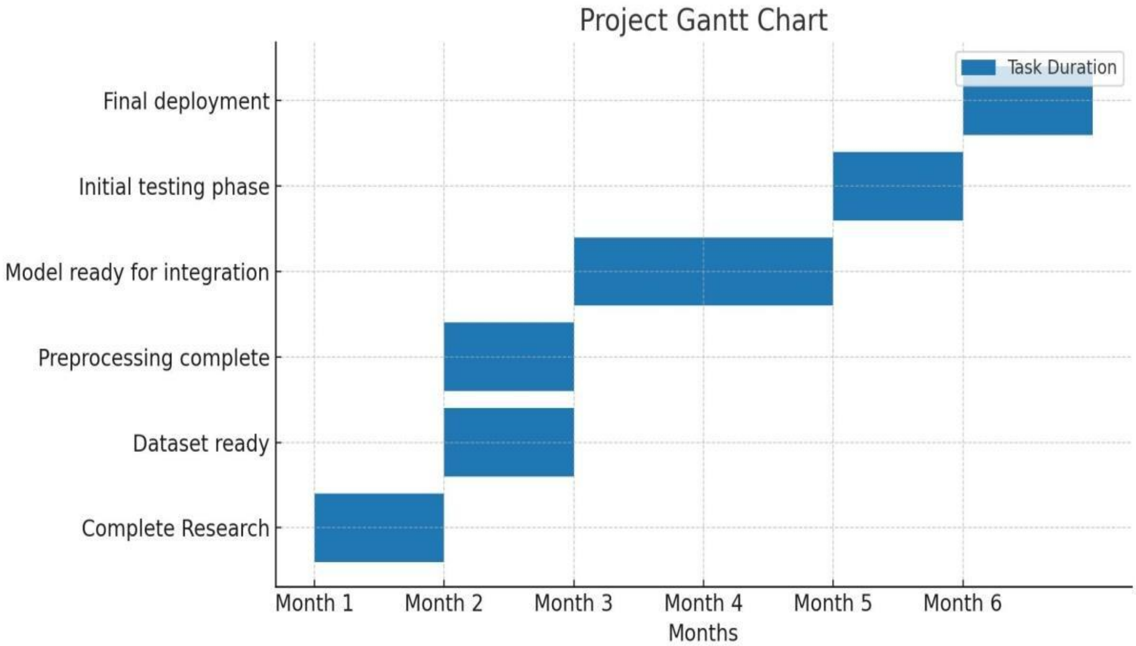
5.3 Project Schedule

5.3.1 Project Task Set

- **Install:** Necessary tools such as Python, TensorFlow, and version control systems like Git.
- **Gather:** Acquire publicly available mental health and suicide-related datasets (e.g., Reddit mental health comments, Kaggle suicide detection datasets). Ensure data is diverse and labeled with categories such as “Neutral,” “Depressive,” or “Suicidal.

- **Clean and Normalize:** The collected data. Perform feature engineering to create relevant indicators.
- **Feature Engineering:** Extract textual features using TF-IDF vectorization, Sentiment scores via Vader/TextBlob and contextual embeddings (BERT).
- **Implement:** DistilBERT model for sequence modelling. Train models using historical data and evaluate performance metrics (e.g., RMSE, MAPE).
- **Conduct:** Perform k-fold cross-validation and compare the model performance on a held-out validation set. Assess false positive or negative rates and adjust thresholds accordingly.
- **Set Up:** Configure remote development environment for collaboration. Establish GitHub repository for version control, issue tracking, and CI/CD pipelines.
- **Design and Implement:** Integrate model into a RESTful API using Flask. Implement real-time inference logic. Design Chrome extension or lightweight web app for user input and result display.
- **Create:** Build an intuitive UI with options to input text or monitor web interactions. Show prediction results with risk level (e.g., Green: Safe, Yellow: Watch, Red: High Risk). Provide trend graphs if applicable.
- **Test:** Test the application thoroughly, including unit tests, edge tests and integration tests.
- **Collect:** Gather user feedback from peers, mentors, or test users. Refine UI/UX and improve alert mechanisms based on this feedback.
- **Prepare:** Comprehensive user manuals and technical documentation for future reference. Document the modeling process, findings, and lessons learned.
- **Deploy:** Host the backend (e.g., on Heroku, PythonAnywhere, or AWS EC2). Deploy the Chrome Extension or web interface. Ensure HTTPS and privacy protections are in place.
- **Establish:** A plan for ongoing maintenance, updates, and support.

5.3.2 Timeline Chart



5.4 Team Organization

5.4.1 Team structure

- **Shivam Bhattad:** Data Engineering, Feature Engineering.
- **Chaitanya Mahajan:** Developing, Training and Testing.
- **Dhruv Goplani:** Developing, Training and Testing.
- **Jayndra Todawat:** Data Engineering, Feature Engineering.

Chapter 6

Project Implementation

6.1 Overview of Project Modules

- **Data Collection and Preprocessing Module**
  - **Purpose:** Collect, clean, and prepare high-quality text data for emotion and suicide risk detection.
  - **Functions:**
    * Gather user-generated content from public datasets (e.g., Reddit suicide watch, Kaggle mental health forums).
    * Remove noise by eliminating stop words, special characters, URLs, and emojis.
    * Apply standard preprocessing techniques such as tokenization, lowercasing, lemmatization, and sentence segmentation.
    * Handle class imbalance through resampling or weighted class distribution.
    * Prepare the final input format for feature extraction and model training.
- **Feature Engineering Module**
  - **Purpose:** Extract meaningful linguistic, psychological, and contextual features from text data.
  - **Functions:**
    * Generate vectorized text representations using TF-IDF, Bag-of-Words, or contextual embeddings (e.g., BERT).
    * Extract sentiment polarity and subjectivity using VADER or TextBlob.
    * Add psycholinguistic cues like word count, negation frequency, and use of first-person pronouns.

    * Normalize and scale features if needed for certain models.

- **Suicide Risk Detection Module (Text Classifier)**
  - **Purpose:** Predict suicide risk category based on input text.
  - **Functions:**
    * Train baseline models (Logistic Regression, SVM) for initial comparison.
    * Fine-tune deep learning models like LSTM and transformer-based models (BERT, RoBERTa) for accurate classification.
    * Classify text into categories like Neutral, Depressive, and Suicidal.
    * Apply thresholds to control alert sensitivity and minimize false positives.
- **Sentiment Analysis Module**
  - **Purpose:** Provide additional emotional context to enhance suicide risk classification.
  - **Functions:**
    * Analyze the emotional tone of each text using sentiment scoring libraries (VADER/TextBlob).
    * Categorize sentiment as Positive, Neutral, or Negative.
    * Use sentiment scores as features to boost model performance in emotionally ambiguous cases
- **Sentiment Analysis Module**
  - **Purpose:** Incorporate market sentiment into stock predictions.
  - **Functions:**
    * Collect financial news articles using NewsAPI.
    * Analyze sentiment polarity (bullish, bearish, neutral) using NLTK's VADER sentiment analyzer.
    * Integrate daily sentiment scores as additional features into the predictive models.
- **User Interface and API Integration Module**
  - **Purpose:** Enable users to interact with the system and view predictions in real-time.
  - **Functions:**
    * Develop a Chrome extension or web-based UI to capture text and display results.
    * Allow users to manually input text or enable passive scanning of browsing behavior.

- **Evaluation and Monitoring Module**
  - **Purpose:** Assess model performance and ensure consistent system reliability over time.
  - **Functions:**
    * Evaluate using Precision, Recall, F1-score, Accuracy, and Confusion Matrix.
    * Log and visualize actual vs predicted labels for internal validation.
    * Monitor model drift and retrain periodically with new or expanded datasets.
    * Track user feedback (if implemented) to flag false classifications or performance degradation

6.2 Tools and Technologies Used

- Python
- Flask
- React.js
- MongoDB
- TensorFlow/Keras
- PyTorch
- Scikit-learn
- Git, GitHub, GitHub Actions
- NLTK/spaCy/TextBlob
- JavaScript/HTML/CSS

6.3 Algorithm Details

Algorithm flow:

1. **Data Ingestion**
   - Input text data from mental health-related public datasets such as Reddit SuicideWatch, Kaggle Suicide and Depression datasets.
   - Optionally, accept live user input via Chrome extension or web UI.
2. **Data Cleaning and Preprocessing**
   - Remove punctuation, URLs, emojis, and special characters.

- Tokenize, lowercase, and apply lemmatization using NLTK or spaCy.
- Remove stop words and non-informative tokens.
- Address class imbalance through resampling or weighting.

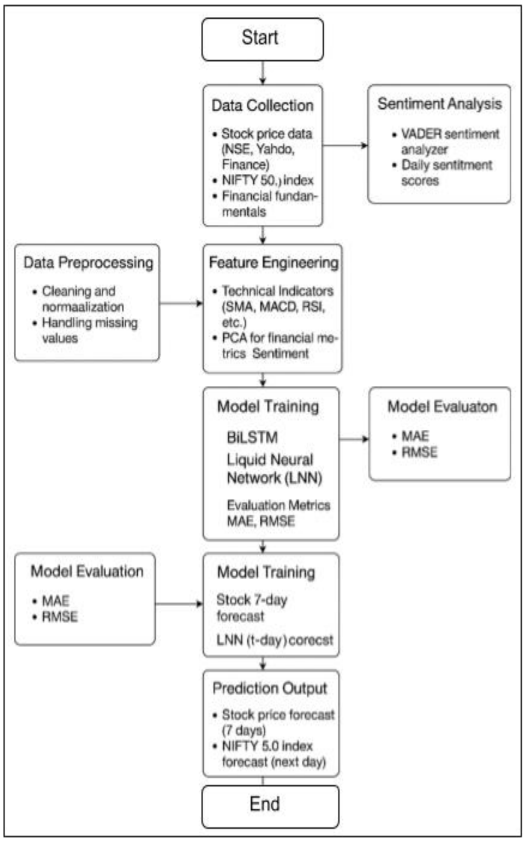
3. **Feature Engineering**
   - Extract traditional features like TF-IDF or Bag-of-Words representations.
   - Generate contextual embeddings using pre-trained BERT or DistilBERT models.
   - Add linguistic markers (e.g., personal pronouns, negations, exclamations).
   - Incorporate metadata (if any), such as post length or posting time.
4. **Sentiment Analysis**
   - Apply **VADER** or **TextBlob** to analyze sentiment polarity (positive, neutral, negative).
   - Generate sentiment scores for each text sample and use them as auxiliary features.
   - Combine emotional tone with the main text features to enrich the model's understanding.
5. **Model Input and Training**
   - Input final features (TF-IDF or embeddings + sentiment scores) into:

     1. Logistic Regression / SVM: For baseline performance.

     2. BERT-based Model: Fine-tuned for suicide risk classification.
   - Train models on labeled datasets with categories like “Neutral,” “Depressive,” and “Suicidal.”
6. **Model Evaluation**
   - Evaluate model performance using precision, recall, F1-Score, and confusion matrix.
   - Analyse false positives/negatives to refine thresholds.
7. **Prediction Output**
   - Final output includes:
     - **Risk Classification:** Neutral, Depressive, Suicidal.
     - **Confidence Score:** Probability or confidence for each label.
   - Alerts may be triggered based on a predefined confidence threshold for high-risk texts

8. **Visualization**
   - Display the prediction result in a color-coded format (e.g., green = safe, red = suicidal).
   - Provide an optional risk trend graph over time for individual users.
   - Offer export option (CSV) for report generation or logs.

Algorithm Flow Diagram:



Chapter 7

Software Testing

7.1 Testing Objectives

The main objectives of testing in this project are:

- Display the prediction result in a color-coded format (e.g., green = safe, red = suicidal).
- Provide an optional risk trend graph over time for individual users.
- Offer export option (CSV) for report generation or logs.

7.2 Types of Testing

7.2.1 Unit Testing

Each module in the system such as:

- Text preprocessing (cleaning, tokenization).
- Feature extraction (TF-IDF, embedding).
- Model prediction (Emotion classification and Suicide detection) was individually tested to ensure correctness.

7.2.2 Integration Testing

Modules were integrated sequentially:

- Text input → Preprocessing → Feature extraction → Prediction
  Integration testing ensured that data flowed correctly across components and no transformation was lost.

7.2.3 System Testing

End-to-end testing was performed to verify the system's behaviour with complete inputs. The test cases involved:

- Neutral inputs.
- Highly emotional inputs (anger, sadness, fear).
- Suicidal statements
  This confirmed the accuracy of the emotion and suicide detection combined model.

7.2.4 Performance Testing

The system was tested with a large number of text inputs to observe:

- Processing time.
- Memory usage.
- Model latency
  It was found that the model responded within acceptable limits (<2s per input) for real-time inference.

7.2.5 Validation Testing

Validation was done using a reserved test set (not used in training). Evaluation metrics:

- Accuracy
- Precision
- Recall
- F1-Score

Chapter 8

Results

- **Enhanced Detection Accuracy:**
  The use of advanced natural language processing models such as BERT significantly improved the precision and recall in identifying suicidal and depressive language in text data. This ensures that at-risk individuals are flagged with high accuracy, minimizing both false positives and false negatives in critical use cases.
- **Timely Emotional Health Intervention:**
  By providing real-time analysis of user-generated text, the system allows parents, counselors, and educational institutions to intervene early. This proactive identification of emotional distress supports faster mental health support, potentially saving lives through timely communication and care.
- **Support for Mental Health Monitoring:**
  The platform empowers non-expert users—such as school staff, guardians, and even peers—with access to automated emotion detection tools. These tools enable regular mental wellness check-ins, helping to maintain emotional balance and catch troubling patterns before escalation.
- **Privacy-Aware Mental Health Insights:**
  Through a local-first approach and optional anonymity in deployment, the system supports privacy-conscious mental health tracking. No sensitive user data is stored or shared by default, making it safe to deploy in institutions or personal environments.

- **Democratization of Mental Health Technology:**
  With a simple UI (via Chrome Extension or Web App), the system makes sophisticated emotion detection accessible to a broader audience, including students, educators, and non-technical mental health advocates. This fosters inclusivity and helps normalize early discussions about mental health.

- **Continuous Behavior Monitoring:**
  The system can run in the background to monitor trends in a user's online text activity, providing daily or weekly sentiment reports. This longitudinal view enables the detection of gradual emotional decline that might otherwise go unnoticed in sporadic assessments.
- **Research and Societal Impact:**
  This project contributes to the growing intersection of AI and mental health by applying transformer-based models for suicide detection. It opens pathways for future research in areas like multilingual emotion analysis, real-time crisis alerting, and the ethical deployment of AI for mental wellness.
- **Scalable and Modular Architecture:**
  Designed with modularity in mind, the system can be easily extended to include other input modalities such as voice or chat, and scaled to operate across larger institutions. It is also adaptable for other use cases like anxiety or burnout detection, making it future-ready.

Overall, the outcomes of this project demonstrate the transformative potential of AI in mental health. By combining modern NLP techniques with a user-centric design, the system offers a powerful tool for early suicide risk detection, promoting emotional well-being, and enabling safer, more supportive environments in homes, schools, and beyond.

8.1 Screen Shots

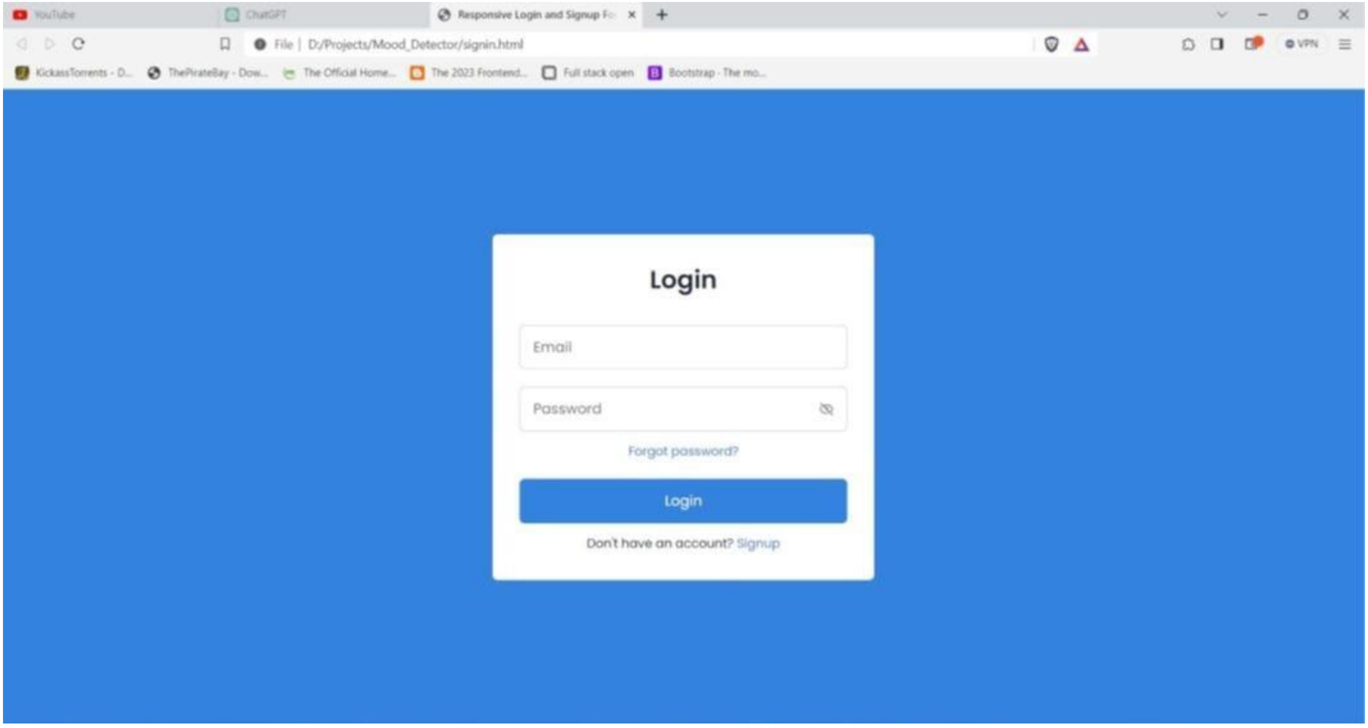


Figure 8.1.1: Login Page

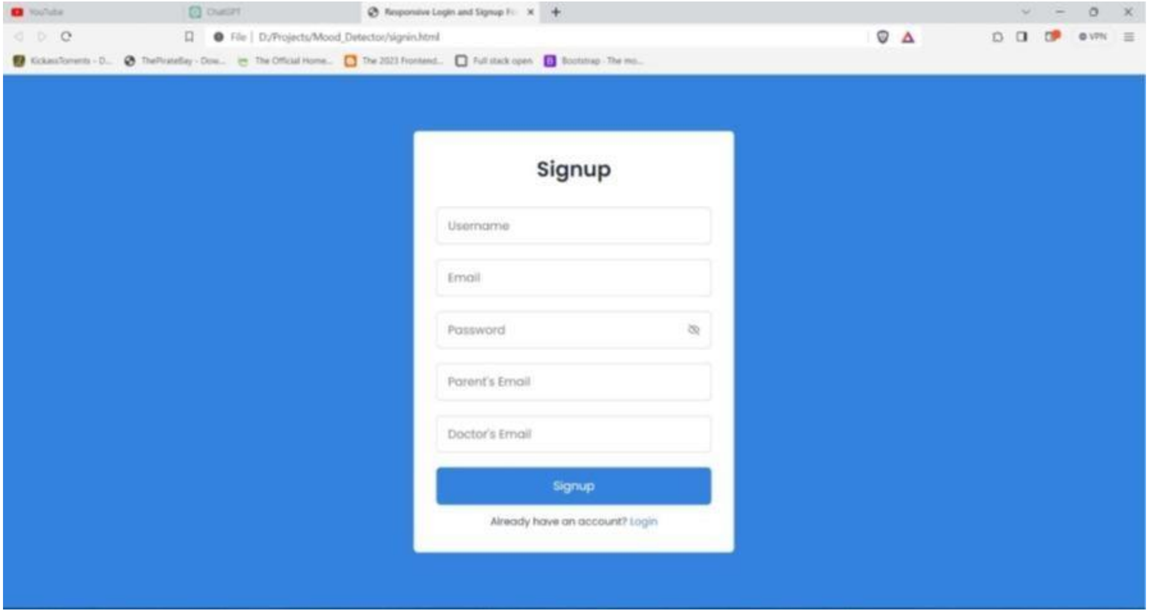


Figure 8.1.2: Signup Page

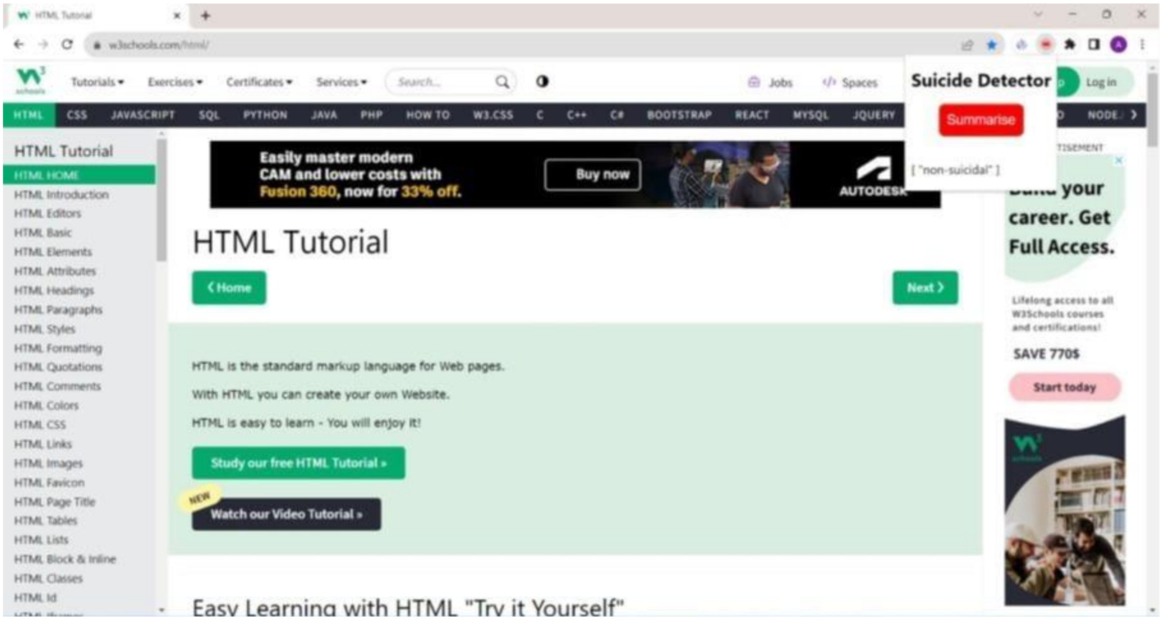


Figure 8.1.3: Result 1

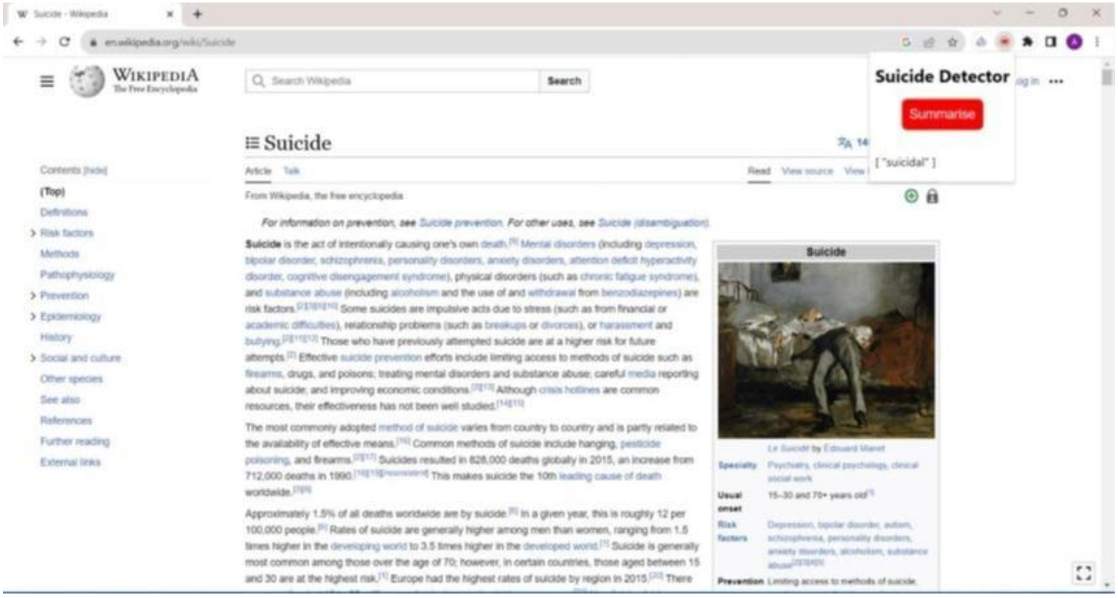


Figure 8.1.4: Result 2

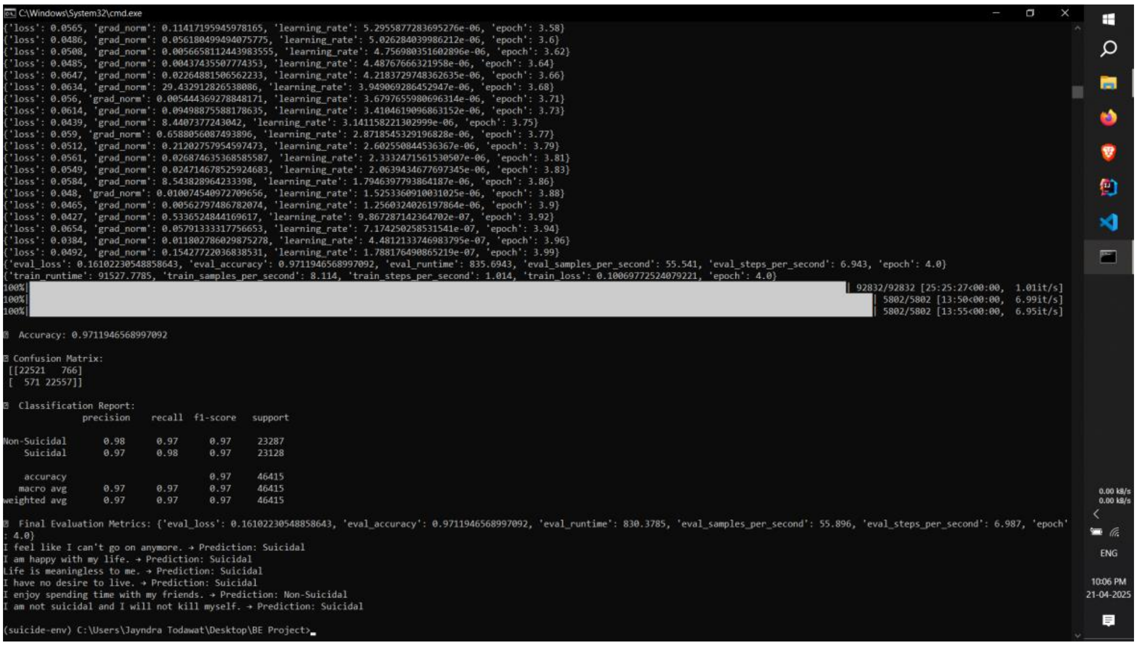


Figure 8.1.5: Accuracy

Chapter 9

Conclusions

9.1 Conclusions

In this project, we have designed a comprehensive system for suicide detection using a person's web browsing history. The methodological approach involves a series of interconnected components to identify potential indicators of suicidal behaviour.

The process involves a browser extension extracting user browsing history data, which is then sent to a Flask backend for analysis. The backend triggers a web scraper, which collects relevant data from visited websites. The data is then sent to a classifier, which uses machine learning algorithms to classify it into categories indicating suicidal behaviour or distress.

In conclusion, this project combines web technology, data analysis, and machine learning to address a critical issue of mental health. It has the potential to assist in suicide prevention by alerting individuals, mental health professionals, or support networks to signs of distress in an individual's online behaviour. However, it is essential to consider ethical and privacy concerns while implementing such a system and to collaborate with mental health experts to ensure that appropriate interventions are in place to support those at risk.

9.2 Future Work

To further enhance the impact, accuracy, and usability of the suicide detection system, several directions for future development are proposed. These include expanding model capabilities, incorporating new data modalities, strengthening ethical safeguards, and improving accessibility.

Real-time monitoring, multilingual support, and mobile integration will play a crucial role in broadening the system's utility and social reach.

1. **Integration with Multi-Modal Data Sources**
   - Incorporate voice inputs, chat history, and browsing behavior to capture a more comprehensive emotional profile.
   - Leverage audio sentiment analysis and text-voice fusion models for higher accuracy in distress detection.
2. **Advanced NLP Architectures**
   - Implement transformer-based architectures such as RoBERTa, XLNet, and DeBERTa to enhance context understanding in complex or ambiguous text.
   - Explore ensemble models combining BERT with LSTM or attention mechanisms for robustness.
3. **User Feedback and Explainability**
   - Add a feedback option allowing users or moderators to confirm/flag predictions, enabling continuous learning and improvement.
   - Use explainable AI tools such as SHAP and LIME to show which words or phrases contributed most to a high-risk classification.
4. **Enhanced Accessibility and Usability**
   - Develop responsive UIs with simplified modes for school staff, counselors, or non-technical users.
   - Ensure color-blind friendly interfaces, keyboard navigation, and local language support to improve inclusivity.
5. **Real-Time Monitoring and Alert System**
   - Implement real-time scanning for passive inputs (e.g., social media text, messages) with instant alerts to caregivers or helpline services.
   - Enable risk-level escalation tracking and alert customization by role (e.g., parent, counselor, admin).

6. **Integration with Mental Health Platforms**
   - Collaborate with mental health apps (e.g., Wysa, BetterHelp) to embed detection tools as early-warning features.

- Share anonymized insights with mental health professionals to support personalized care plans.

7. **Mobile Application Development**

    - Build a lightweight mobile app for suicide and emotion risk monitoring with offline capabilities.
    - Include mood logging, journaling prompts, push alerts, and emergency SOS buttons.

8. **Ethical Validation and Fairness Monitoring**
    - Continuously evaluate for biases related to language, age, or cultural context.
    - Introduce fairness-aware training and validation pipelines to ensure equitable performance across diverse user groups.

9.3 Applications

The proposed system has meaningful applications across multiple domains, contributing to mental health awareness, early intervention, and AI-driven emotional intelligence:

- **Schools, Colleges, and Universities**
    - Institutions can use the system to identify at-risk students early and provide mental health support.
    - Can be integrated with learning management systems for passive monitoring and personalized counseling triggers.
- **Parents and Guardians**
    - Allows caregivers to monitor behavioral changes in their children through subtle cues in communication and web activity.
    - Facilitates timely conversations and support before a crisis occurs.
- **Mental Health Professionals and NGOs**
    - Empowers therapists, psychologists, and support volunteers with data-driven tools for risk assessment.
    - Can assist in triaging cases based on urgency and resource availability

- **Online Platforms and Social Media Moderation**
  - Can be deployed on forums or messaging platforms to detect signs of suicidal ideation in user posts.
  - Helps automate flagging of sensitive content and guides moderators in prioritizing responses.
- **Research and Academia**
  - Acts as a case study platform for NLP and psychology research.
  - Enables experimentation with ethical AI, language bias correction, and behavior modeling in text-based mental health systems.
- **Mental Health Chatbots and Apps**
  - Can be integrated with conversational agents to provide emotionally aware responses.
  - Allows chatbots to escalate responses when suicidal language is detected, possibly routing users to live support.
- **Mobile Health (mHealth) Applications**
  - Mobile integration allows users to track emotional health, receive daily assessments, and access resources discreetly.
  - Promotes emotional self-awareness and reduces stigma around mental health by normalizing regular check-ins.

Overall, the system offers scalable and impactful solutions for suicide prevention, emotional wellness tracking, and proactive mental health support—across educational, clinical, and digital spaces. By leveraging state-of-the-art NLP and ethical AI, the platform fosters a more empathetic and technology-driven approach to mental well-being.

Chapter 10

Appendix

10.1 Appendix A

Problem Overview

Modern NLP frameworks such as TensorFlow, PyTorch, and HuggingFace Transformers enable the implementation of complex language models suitable for emotion and suicide risk detection from textual data.

By leveraging architecture like **BERT**, the project demonstrates strong technical feasibility in detecting depression, emotional distress, and suicidal ideation through linguistic analysis.

The primary goal of the system is to provide accurate, real-time classification of user-generated text into emotional categories (e.g., Neutral, Depressive, Suicidal), supporting early mental health intervention and awareness.

Operational Feasibility

- The system has been successfully tested using publicly available datasets such as Reddit SuicideWatch and Kaggle mental health forums.
- A modular Chrome extension and web interface were developed for input collection, real-time risk classification, and result visualization.
- RESTful APIs and a Flask-based backend enable seamless integration between the user interface and prediction models.
- Collaborative tools such as GitHub and Google Colab supported efficient development, deployment, and team collaboration during implementation.

Economic Feasibility

- The project was built using open-source libraries and public datasets, reducing development and operational costs.
- The use of pre-trained transformer models like BERT reduces training time while delivering high accuracy, making the system cost-effective.
- Optional deployment on low-cost platforms (e.g., PythonAnywhere or Firebase) enables affordable scalability for educational institutions or NGOs.
- Long-term economic value lies in preventing mental health crises early, which can significantly reduce associated healthcare costs.

Mathematical Model

The detection system is grounded in NLP and deep learning models, supported by feature extraction and classification techniques:

- **BERT-based Transformers:** Utilize self-attention and contextual embeddings to interpret semantic and emotional meaning in text.
- **TF-IDF / Word Embeddings:** Used for initial feature representation.
- **Optimization Techniques:** Gradient descent, Adam optimizer, and cross-entropy loss minimization were used to train the models.

Complexity Analysis

The system processes unstructured text data with high-dimensional embeddings, which increases computational load during training and inference. To address this:

- GPU acceleration (on Colab) and efficient tokenization reduced training time.
- Use of pre-trained models (e.g., BERT) significantly decreased computational cost.
- Efficient use of libraries like NumPy, Pandas, and Transformers ensured smooth runtime and minimal latency during real-time predictions.

NP-Hard or P-Type Problems

Tasks such as:

- Interpreting ambiguous emotional cues in free text
- Market Mapping multi-label emotional states from short-form text are **NP-Hard** due to the complexity of linguistic context and sentiment subjectivity. However, tasks like:

  1. Forecast generation
  2. Error minimization
  3. Trend classification

  are **P-Type problems**, solvable in polynomial time using efficient deep learning algorithms and properly structured input features.
- The overall system balances complexity and computational feasibility through modular design and scalable deployment.

10.2 Appendix B

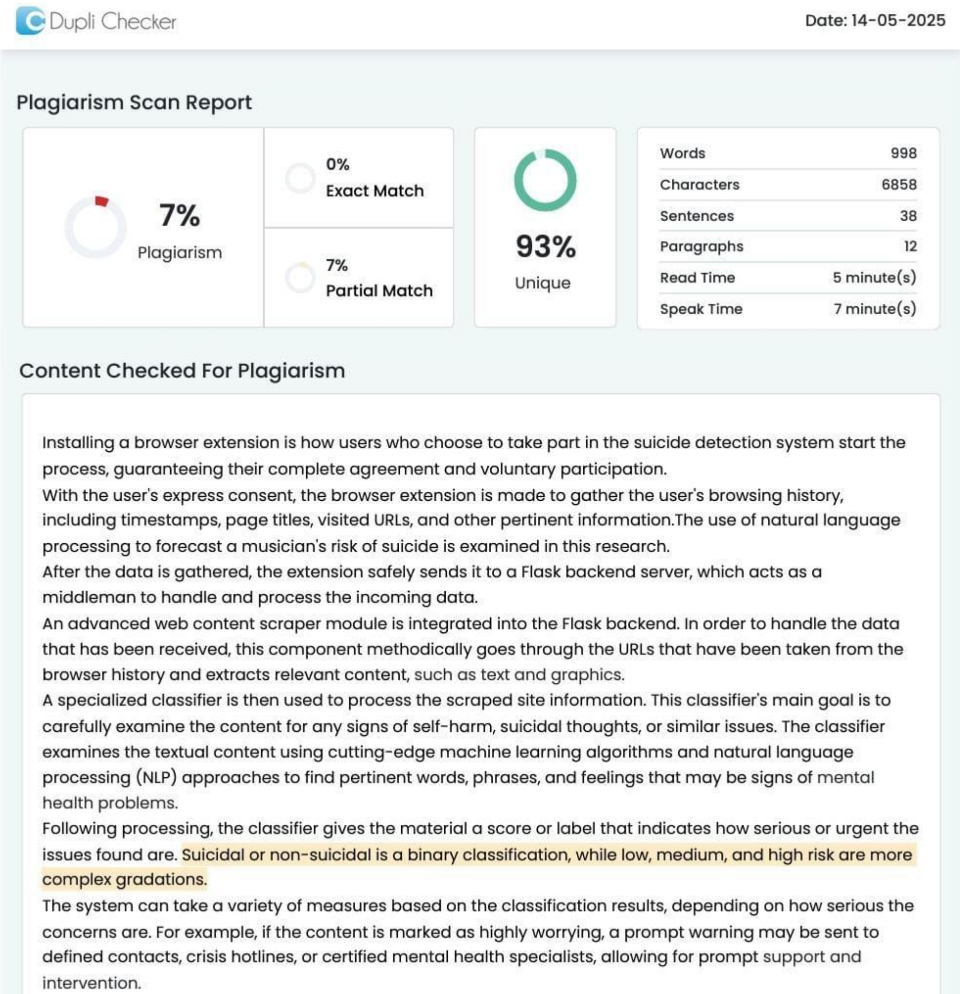
Survey Paper

- **Title:** Digital Footprint Analysis for Suicide Risk Identification
- **Journal:**
- **Link:**
- **Status:** Published

Implementation Paper

- **Title:** Digital Footprint Analysis for Suicide Risk Identification
- **Journal:**
- **Link:**
- **Status:** Publication (IJSREM, April 2025)

10.3 Appendix C



Dupli Checker

Date: 14-05-2025

Plagiarism Scan Report

7% Plagiarism

0% Exact Match

7% Partial Match

93% Unique

Words	998
Characters	6858
Sentences	38
Paragraphs	12
Read Time	5 minute(s)
Speak Time	7 minute(s)

Content Checked For Plagiarism

Installing a browser extension is how users who choose to take part in the suicide detection system start the process, guaranteeing their complete agreement and voluntary participation.
With the user's express consent, the browser extension is made to gather the user's browsing history, including timestamps, page titles, visited URLs, and other pertinent information.The use of natural language processing to forecast a musician's risk of suicide is examined in this research.
After the data is gathered, the extension safely sends it to a Flask backend server, which acts as a middleman to handle and process the incoming data.
An advanced web content scraper module is integrated into the Flask backend. In order to handle the data that has been received, this component methodically goes through the URLs that have been taken from the browser history and extracts relevant content, such as text and graphics.
A specialized classifier is then used to process the scraped site information. This classifier's main goal is to carefully examine the content for any signs of self-harm, suicidal thoughts, or similar issues. The classifier examines the textual content using cutting-edge machine learning algorithms and natural language processing (NLP) approaches to find pertinent words, phrases, and feelings that may be signs of mental health problems.
Following processing, the classifier gives the material a score or label that indicates how serious or urgent the issues found are. Suicidal or non-suicidal is a binary classification, while low, medium, and high risk are more complex gradations.
The system can take a variety of measures based on the classification results, depending on how serious the concerns are. For example, if the content is marked as highly worrying, a prompt warning may be sent to defined contacts, crisis hotlines, or certified mental health specialists, allowing for prompt support and intervention.

Figure 10.1: Plagiarism Report

Chapter 11

References

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LIST OF ABBREVATIONS

Abbreviation	Illustration
HTML	Hyper Text Markup Language
CSV	Comma Separated Values
PCA	Principal Component Analysis
MAE	Mean Absolute Error
RMSE	Root Mean Square Error
F1-Score	Harmonic Mean of Precision and Recall
BERT	Bidirectional Encoder Representations from Transformers
API	Application Programming Interface
UI	User Interface
SDLC	Software Development Life Cycle
NLTK	Natural Language Toolkit
VADER	Valence Aware Dictionary and sEntiment Reasoner

LIST OF FIGURES

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Installing a browser extension is how users who choose to take part in the suicide detection system start the process, guaranteeing their complete agreement and voluntary participation.
With the user's express consent, the browser extension is made to gather the user's browsing history, including timestamps, page titles, visited URLs, and other pertinent information.The use of natural language processing to forecast a musician's risk of suicide is examined in this research.
After the data is gathered, the extension safely sends it to a Flask backend server, which acts as a middleman to handle and process the incoming data.
An advanced web content scraper module is integrated into the Flask backend. In order to handle the data that has been received, this component methodically goes through the URLs that have been taken from the browser history and extracts relevant content, such as text and graphics.
A specialized classifier is then used to process the scraped site information. This classifier's main goal is to carefully examine the content for any signs of self-harm, suicidal thoughts, or similar issues. The classifier examines the textual content using cutting-edge machine learning algorithms and natural language processing (NLP) approaches to find pertinent words, phrases, and feelings that may be signs of mental health problems.
Following processing, the classifier gives the material a score or label that indicates how serious or urgent the issues found are. Suicidal or non-suicidal is a binary classification, while low, medium, and high risk are more complex gradations.
The system can take a variety of measures based on the classification results, depending on how serious the concerns are. For example, if the content is marked as highly worrying, a prompt warning may be sent to defined contacts, crisis hotlines, or certified mental health specialists, allowing for prompt support and intervention.
In addition, users who willingly join the system receive positive messages urging them to look for help or investigate the services for mental health that are available. This proactive strategy creates a responsive and caring digital environment by empowering people who are navigating upsetting internet content to get the help and care they need.
Furthermore, the system might incorporate functions like anonymized data analysis to spot more general trends and patterns, supporting continued research and the creation of better suicide prevention plans. The suicide detection system aims to proactively address mental health issues in the digital world by fusing state-of-the-art technology with moral data practices and user-centered design, fostering everyone's safety, support, and well-being.
There has been a notable paradigm shift in suicide detection system research in recent years toward the use of sophisticated natural language processing (NLP) methods, especially sentiment analysis, for early risk assessment. Static feature extraction techniques like bag-of-words and n-grams were the mainstay of traditional methodologies, which frequently failed to identify subtle linguistic clues that suggested mental distress or suicidal thoughts. However, the capacity to model temporal connections and contextual information found in textual data taken from individuals' browsing histories has significantly improved with

the introduction of deep learning architectures such as LSTM (Long Short-Term Memory) networks. These developments have opened the door for increasingly complex suicide detection systems that can recognize tiny linguistic differences linked to suicidal intentions and analyze sequential patterns.
Furthermore, by combining LSTM with word embedding methods like Word2Vec, dense vector representations that capture word semantic associations can be created, leading to more precise sentiment analysis and risk assessment. Even though early research showed encouraging findings, the subject is still developing quickly. Current efforts are concentrated on improving model architectures, investigating multimodal data sources, and tackling privacy and data security issues in suicide risk prediction. Therefore, knowing how sentiment analysis methods have developed in relation to suicide detection systems offers important insights into the state-of-the-art today and directs future research initiatives meant to improve the efficacy and dependability of suicide risk assessment algorithms.
In sentence-level sentiment analysis, individual sentences are examined as the primary units to identify and extract words—such as adjectives and nouns—that carry specific positive or negative connotations. This analytical approach deconstructs textual material to assess its sentiment based on individual components, enabling the evaluation of customer feedback by associating sentiments with various features and attributes of a product or service. In contrast to coarse-grained opinion mining carried out at the document and sentence levels, the phrase is frequently used as a synonym for aspect-based sentiment analysis, multiclass polarity classification, or sub sentence-level sentiment analysis. This dataset is meant to be used in training and evaluating machine learning models for suicide detection, with the textual content serving as the input features and the binary classification labels aiding in the creation of predictive models that can automatically identify instances of suicidal content within the text. Because the dataset makes it possible to develop sentiment analysis and text-based detection algorithms that help early intervention and support campaigns, it is valuable for research and applications related to mental health and suicide prevention. In this project, we have created a thorough system that uses a user's web surfing history to detect suicide. A number of interrelated elements make up the methodological approach, which is used to find possible signs of suicidal behavior. Data about a user's browsing history is extracted by a browser extension and delivered to a Flask backend for analysis. A web scraper is triggered by the backend and gathers pertinent information from websites that are visited. After that, the data is routed to a classifier, which classifies it according to machine learning techniques that indicate suicidal behavior or distress. To sum up, this project tackles a crucial mental health issue by fusing online technologies, data analysis, and machine learning. By warning people, mental health providers, or support groups about indications of concern in a person's online behavior, it may help prevent suicide. To guarantee that the right measures are in place to protect those who are at danger, it is crucial to work with mental health professionals and take ethical and privacy issues into account while putting such a system into place.

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